Wireless Intrusion Detection and Prevention with Machine Learning

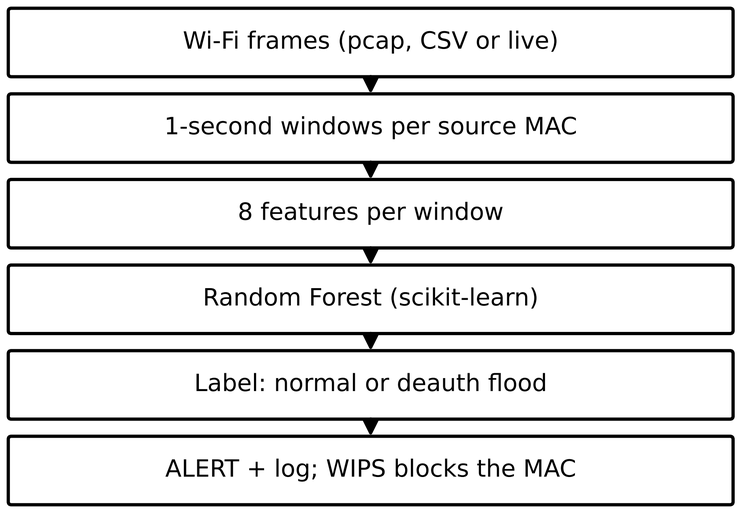
Detecting Wi-Fi deauthentication flood attacks with a Random Forest classifier (WIPS prototype)

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1. Problem and aim

Deauthentication frames are unauthenticated management frames. An attacker can spoof them to repeatedly disconnect clients from an access point, causing denial of service and enabling evil twin and handshake-capture attacks. Aim: detect deauth floods automatically, raise an alert and block the offender.

2. How the system works



3. Features (per MAC, per second)

- Frame count; deauth/disassoc count and ratio
- Management-frame ratio; unique destinations
- Broadcast ratio; signal strength (RSSI) mean and std

4. Data and model

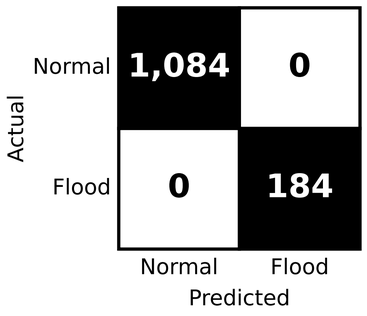
- Dataset: AWID3 Deauth capture (Kaggle copy), files 17-32: 776,472 frames, 38,942 labelled attack.
- Frames grouped per source MAC per second: 4,226 windows.
- Random Forest, 100 trees, depth 8. Time-ordered split: train on the earliest 70% of windows, test on the latest 30%.
- Same code also reads own monitor-mode pcaps and live capture on an authorised lab.

5. Results

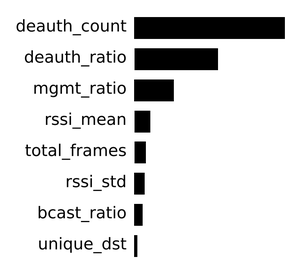
Accuracy: 100.0%

1,268 unseen test windows. Attack-class precision 1.00, recall 1.00, F1 1.00. No false alarms, no missed attacks.

Confusion matrix



Feature importance



Note: deauth_count is the strongest feature (0.47) because attack frames are deauth frames. Subtler attacks would be harder.

6. Alert and WIPS action

```
[ALERT] Deauth flood from
0c:9d:92:54:fe:34 | 24 deauth frames/s
[WIPS ] 0c:9d:92:54:fe:34 added
to blocked_macs.json
(all alerts also in alerts.log)
```

The detector flagged 383 of 4,226 windows and block-listed 11 source MACs. The list can feed an AP MAC filter or controller.

7. Limitations

- One attack type, one capture. RSSI values are implausibly low (median -127 dBm attack, -116 dBm normal), so signal features are unreliable.
- Spoofed MACs: 11 addresses were block-listed, possibly victim devices, so MAC blocking can hit innocent devices.
- WIPS only records the MAC; it does not yet push it to an AP.

8. What I would improve

- Test on other AWID3 attacks (disassoc, evil twin) and own lab captures.
- Evaluate on a completely separate capture.
- Use an allow-list of known APs; detect spoofing with sequence numbers.
- Push blocks to the AP or controller automatically.

AI use statement

Claude (AI assistant) helped write the detection code, the AWID3 data converter and this poster generator, and helped me debug. I ran and tested everything myself and can explain how the code works.

Ethics

Training used the public AWID3 dataset. I generated no attacks; any own capture is limited to an authorised isolated testbed.